

VibeCast – Microservices-based Collaborative Video Platform

Core Concept

A platform where creators upload videos and teams collaborate using timestamp comments, frame annotations, and live sessions.

Microservices Architecture

- 1 Auth Service – JWT + Refresh Tokens, Redis caching
- 2 User Service – Profiles, roles, team management
- 3 Video Service – Upload, metadata, Cloud storage
- 4 Streaming Service – WebRTC / HLS live streaming
- 5 Comment Service – Timestamped comments
- 6 Annotation Service – Frame drawing & sync
- 7 Notification Service – Real-time alerts via WebSockets
- 8 API Gateway – Central routing

Tech Stack

- 1 Frontend: React (Vite), Redux Toolkit
- 2 Backend: Node.js, Express
- 3 Database: MongoDB
- 4 Cache: Redis
- 5 Media: Cloudinary / AWS S3
- 6 Realtime: Socket.io / WebRTC
- 7 DevOps: Docker, Kubernetes

System Flow

- 1 Upload Flow: Frontend → Gateway → Video Service → Cloud Storage → DB
- 2 Comment Flow: User → Comment Service → DB → Redis → WebSocket
- 3 Live Flow: Host → Streaming Service → Participants (WebRTC)

Advanced Features

- 1 AI video summaries & speech-to-text
- 2 Version control for videos
- 3 Role-based access control
- 4 Playback synchronization

Challenges

- 1 Handling large video uploads
- 2 Real-time sync between users
- 3 Low latency streaming
- 4 Scalability of microservices
- 5 Data consistency

Build Roadmap

- 1 Phase 1: Auth + Upload + Playback
- 2 Phase 2: Comments + Teams
- 3 Phase 3: Real-time features
- 4 Phase 4: Live streaming
- 5 Phase 5: Scaling + AI features